

Mischievous Little Guy

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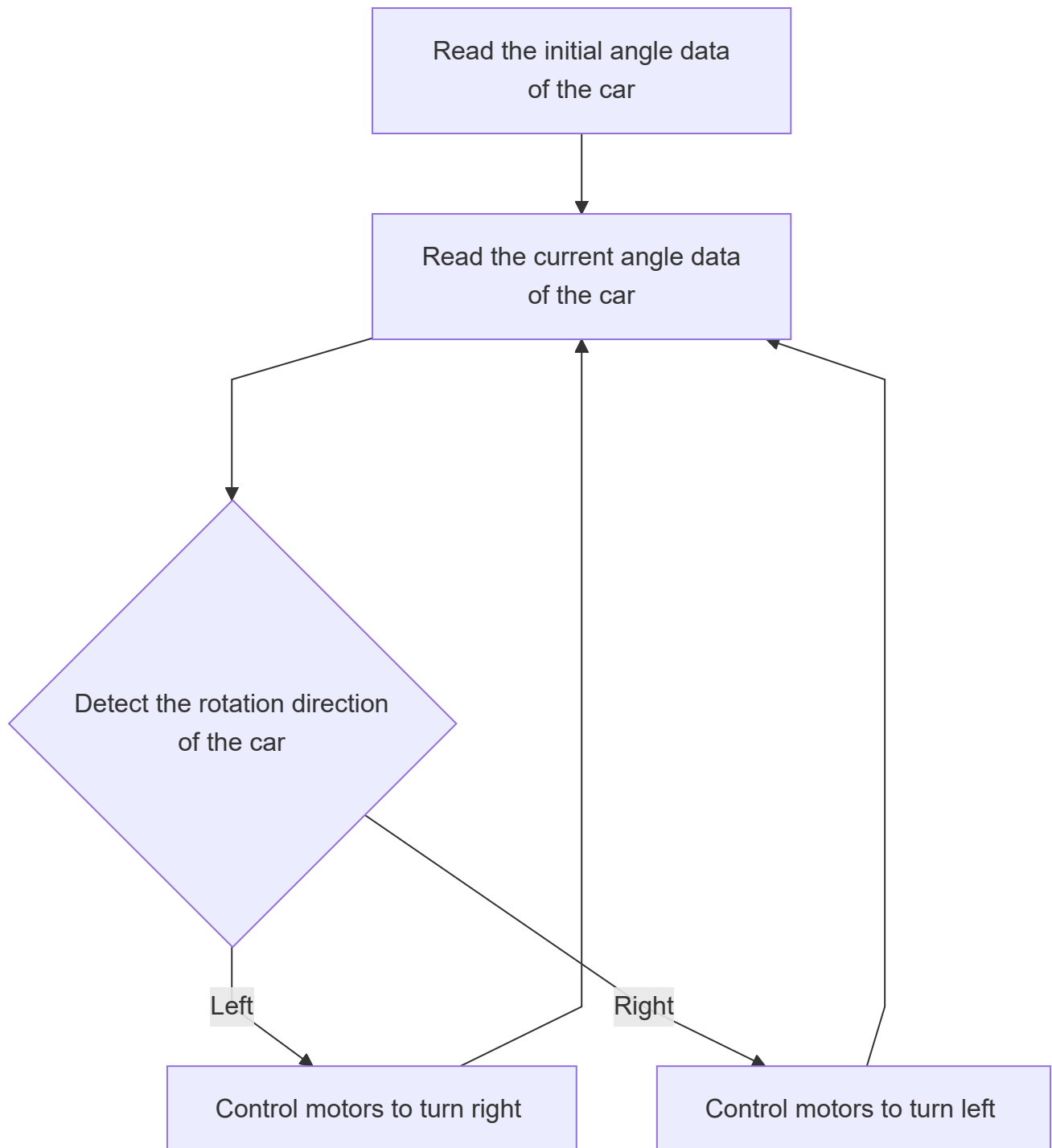
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Function Description

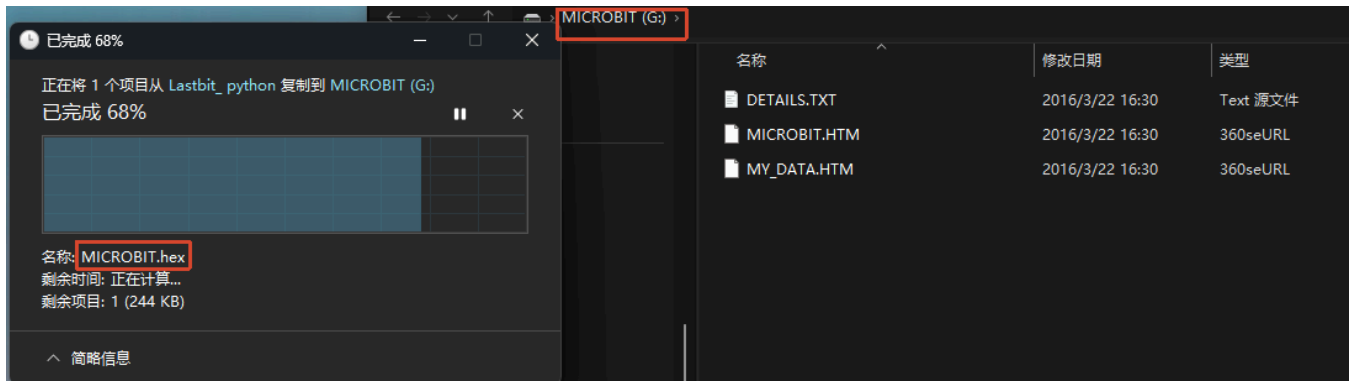
In this section, we will learn to use the compass sensor to implement the "Naughty Little Guy".

Working Principle



Experiment Steps

Make sure `LASTBIT.hex` is flashed before flashing, otherwise an error will occur and the module cannot be imported



1. Before flashing, switch the switch to the OFF position and use a microUSB data cable to connect the computer and the micro:bit main board. **Note that it is the Microbit interface, not the Charging port on the expansion board.**
2. Open the Mu software. Here you can directly copy the program source code we provided; the operation steps are as follows. You can also enter the code manually in the editing window. Note! All English letters and symbols should be entered in English input mode. Use the Tab key for indentation, and the last line should end with a blank program.

Click the **New** button in the toolbar at the top left, and a file titled "untitled" will appear.



3. Copy the content of the **Code Implementation** section provided below into the current file. You will notice a red dot after the title bar, indicating that the code content has been modified but is in an unsaved state.

Whether to save or not does not affect code checking or flashing.

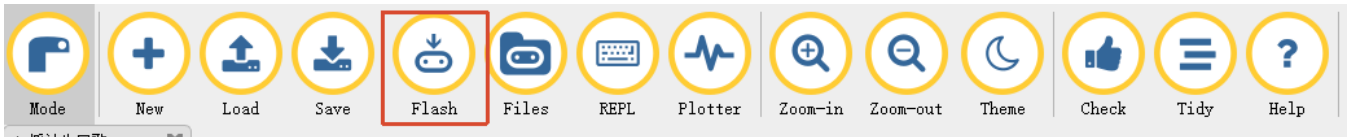
4. Now you can click **check** to verify whether there are any problems with the code format.



5. Now if you have already connected the MICROBIT to the PC in advance following the steps above, and flashed **LASTBIT.hex**, the next step is to download the current program to the Microbit.

Click the **Flash** tool in the toolbar to start flashing!

During flashing, the orange-yellow indicator light near the Microbit main board will blink frequently. When the editor shows **Copied code onto micro:bit.** at the bottom, it means the flashing is complete, and the indicator light will no longer blink after flashing.



6. Unplug the microUSB data cable and switch the switch to the ON position.

Code Implementation

```
# 9. Naughty Little Guy (Naughty Little One)
from microbit import *
from lastbit import LASTBIT

Lastbit = LASTBIT()

# Base speed
base_speed = 150
# Proportional coefficient
KP = 5

# Initialize servos
Lastbit.set_all_servos(90)
Lastbit.set_servo('S2', 150)
sleep(2000)

# Initialize compass (calibration required for first use)
# Prompt: TILT TO FILL SCREEN
# compass.calibrate()

# Get target direction
target_yaw = compass.heading()

display.show(Image.YES)

while True:
    # Get current direction
    new_yaw = compass.heading()

    # Calculate error
    error = new_yaw - target_yaw

    # Handle angle wrap-around at 360 degrees
    if error > 180:
        error -= 360
    if error < -180:
        error += 360

    # Ignore small errors
```

```
if abs(error) < 2:
    error = 0

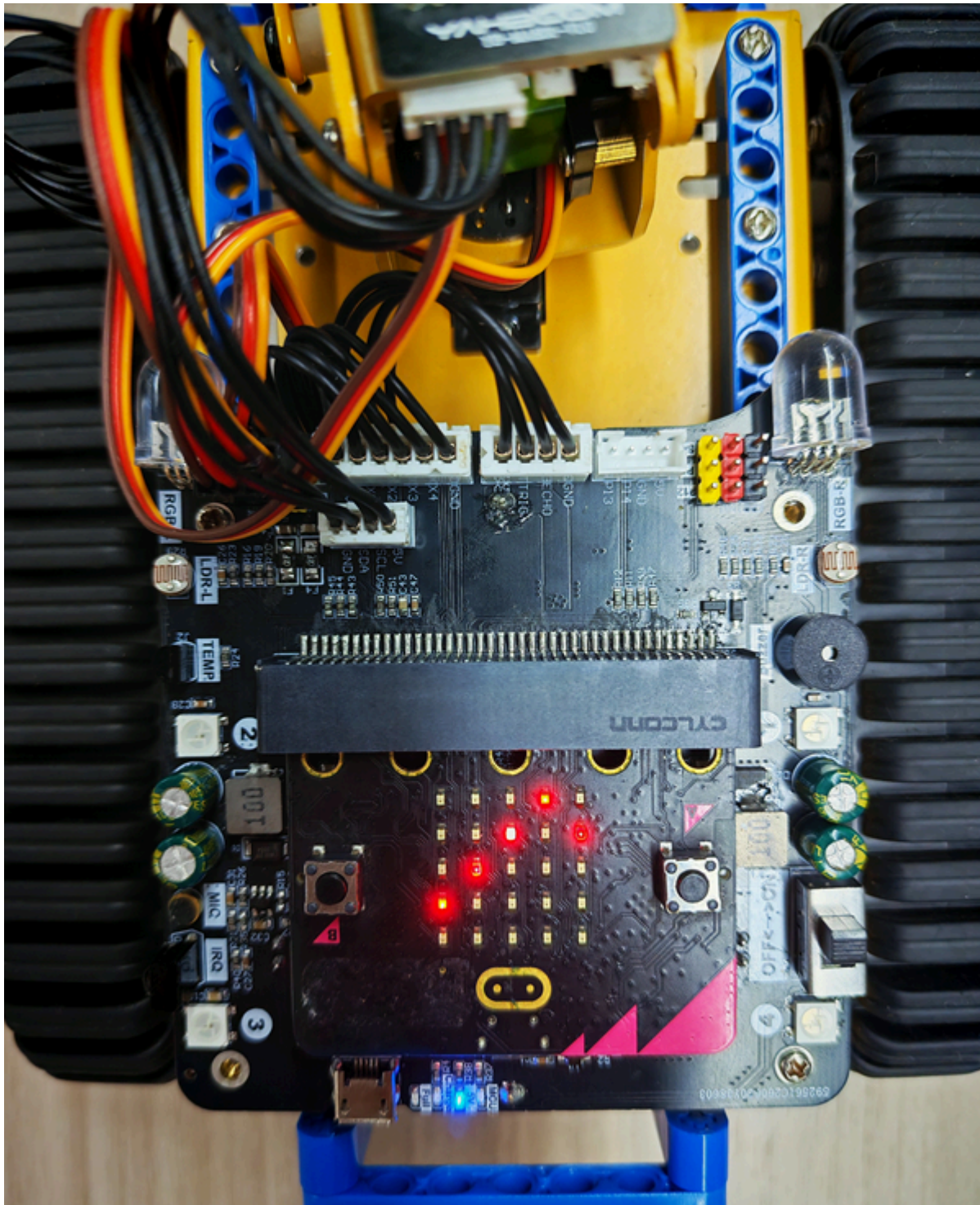
# calculate left and right wheel speeds
left_speed = base_speed - error * KP
right_speed = base_speed + error * KP

# Set motors
Lastbit.set_motor(left_speed, right_speed)

sleep(200)
```

Experimental Phenomenon

After flashing is complete and the power is turned on, the car will first detect the current direction information and display a checkmark "√" on the matrix screen, then move forward. If an external force rotates the car to the left, the car will turn right; if the car is rotated to the right, it will turn left.



Notes

- If the screen scrolls the prompt "TILT TO FILL SCREEN", this is a warning message from the Microbit, which means tilting the device to fill the screen with LEDs. At this time, you need to pick up the car and tilt it in different directions until all its 5x5 LED matrix lights are on, completing sensor calibration. After calibration is complete, a smiley face will be displayed, and then the car can move normally.
- On first use of the compass, you can uncomment `compass.calibrate()` and follow the prompts. After calibration, comment out this calibration function again.